GlucoTwin AI — System Architecture

A synthetic-data proof of concept for a Type 2 diabetes digital twin

1. Historical / EHR stream Synthetic patient profile: HbA1c, BMI, lipids, blood pressure, medication, family history, TCF7L2 marker and diet pattern.	**2. Dynamic time-series stream** Synthetic CGM/glucose, heart rate, HRV, steps, sleep and meal events.
3. Data preparation Patient/time alignment, feature engineering, rolling summaries and validation.	**4. Prediction model** Gradient-boosted tree classifier estimates whether glucose may reach 180 mg/dL within approximately two hours.
5. Digital Twin state Selected synthetic patient, monitoring time and recent signals.	**6. What-if simulator** Compare meal choices, post-meal walking and sleep-duration settings.
7. Doctor-facing dashboard Risk estimate, recent glucose curve, meal markers, health record and risk-associated features.	**8. Evaluation and safeguards** Simulator-based evaluation only. Independent validation is required; not for clinical use.

Data flow: synthetic EHR + dynamic wearable stream → feature engineering → model prediction → virtual-patient visualization and scenario comparison. This is a research prototype, not a clinically validated medical device.